Supervised Learning Assignment

Classifier Comparison with 5×2 Cross-Validation and Friedman Test

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1 Introduction

The goal of this assignment is to compare the performance of different supervised learning classifiers on several two-dimensional datasets and to assess whether any of the classifiers is statistically significantly better than the others. In particular, we focus on:

- Support Vector Machine (SVM) with a linear kernel,
- SVM with a radial basis function (RBF) kernel,
- k-Nearest Neighbors (KNN) classifier.

Performance is estimated using the 5×2 cross-validation procedure, and a Friedman test with Nemenyi post-hoc analysis is applied to the average ranks in order to evaluate the presence of statistically significant differences among classifiers.

2 Datasets

Four synthetic datasets (dataset1–dataset4) are used in the experiments. Each dataset:

- is two-dimensional,
- contains two classes (labels 1 and 2),
- may exhibit different levels of class overlap and geometry.

Figure 1 shows the scatter plots of the four datasets, with different markers and colors used for the two classes.

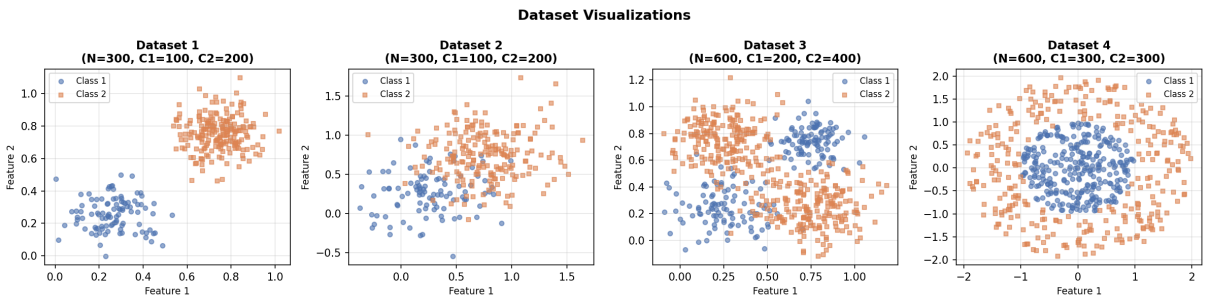

Figure 1: Scatter plots of the four datasets used in the supervised learning assignment.

3 Experimental Setup

For each classifier and each dataset, the performance is estimated using the 5×2 cross-validation scheme:

1. For each of 5 repetitions:
   (a) The dataset is split into two stratified halves (50% training, 50% test per class).
   (b) The classifier is trained on the first half and evaluated on the second half.
   (c) The roles of the halves are reversed: trained on the second half and evaluated on the first half.
   (d) The mean of the two accuracies is stored as the result for that repetition.

2. The average over the 5 repetitions is reported as the 5×2 cross-validation accuracy for that classifier–dataset pair.

This procedure is repeated independently for all four datasets and for each of the three classifiers. The implementation is provided in the Python script `classifier_comparison.py`, which also saves the resulting figures to `fig1_datasets.png` and `fig2_results.png`.

4 Results

4.1 Accuracy Summary

The script computes, for each classifier, a vector of four accuracies corresponding to the four datasets. The obtained 5×2 cross-validation accuracies are reported in Table 1.

Classifier	**DS1**	**DS2**	**DS3**	**DS4**
SVM Linear	1.0000	0.8793	0.6667	0.5780
SVM RBF	1.0000	0.8687	0.9310	0.9753
KNN	1.0000	0.8427	0.9207	0.9610

Table 1: 5×2 cross-validation accuracies for each classifier on the four datasets.

4.2 Friedman Test and Critical Difference Diagram

To compare the classifiers across the four datasets, we applied the Friedman test on the rank matrix. For each data set, classifiers are ranked according to their precision (rank 1 = best). The average rank of each classifier on the four datasets is then computed.

The Friedman test returns a test statistic and a p-value. In this assignment, the resulting p-value is:

$$p \;=\; p_\text{Friedman},$$

as printed by the script. The script also checks whether $p < 0.05$ determines statistical significance at the $\alpha = 0.05$ level.

When the Friedman test does not reject the null hypothesis ($p \geq 0.05$), we conclude that there is no statistically significant difference among the classifiers. In that case, any differences in average ranks are not large enough to be considered significant.

The post-hoc analysis is performed using the Nemenyi test. The critical difference (CD) is given by:

$$\text{CD} = q_\alpha \sqrt{\frac{k(k+1)}{6N}},$$

where k is the number of classifiers ($k = 3$), N is the number of datasets ($N = 4$), and q_α is the critical value from the Nemenyi distribution (for $\alpha = 0.05$ and $k = 3$, $q_\alpha = 2.343$). Substituting these values, we obtain

$$\text{CD} = 2.343\sqrt{\frac{3(3+1)}{6 \cdot 4}} = 2.343\sqrt{\frac{12}{24}} = 2.343\sqrt{0.5} \approx 1.6568.$$

This is exactly the CD value used to draw the horizontal bars in the critical difference diagram.

Figure 2 shows the critical difference diagram produced by the script. Classifiers are placed on the horizontal axis according to their average rank, and a horizontal bar of length equal to the CD is drawn for each classifier. If the bars of two classifiers overlap, their performance is not significantly different according to the Nemenyi test.

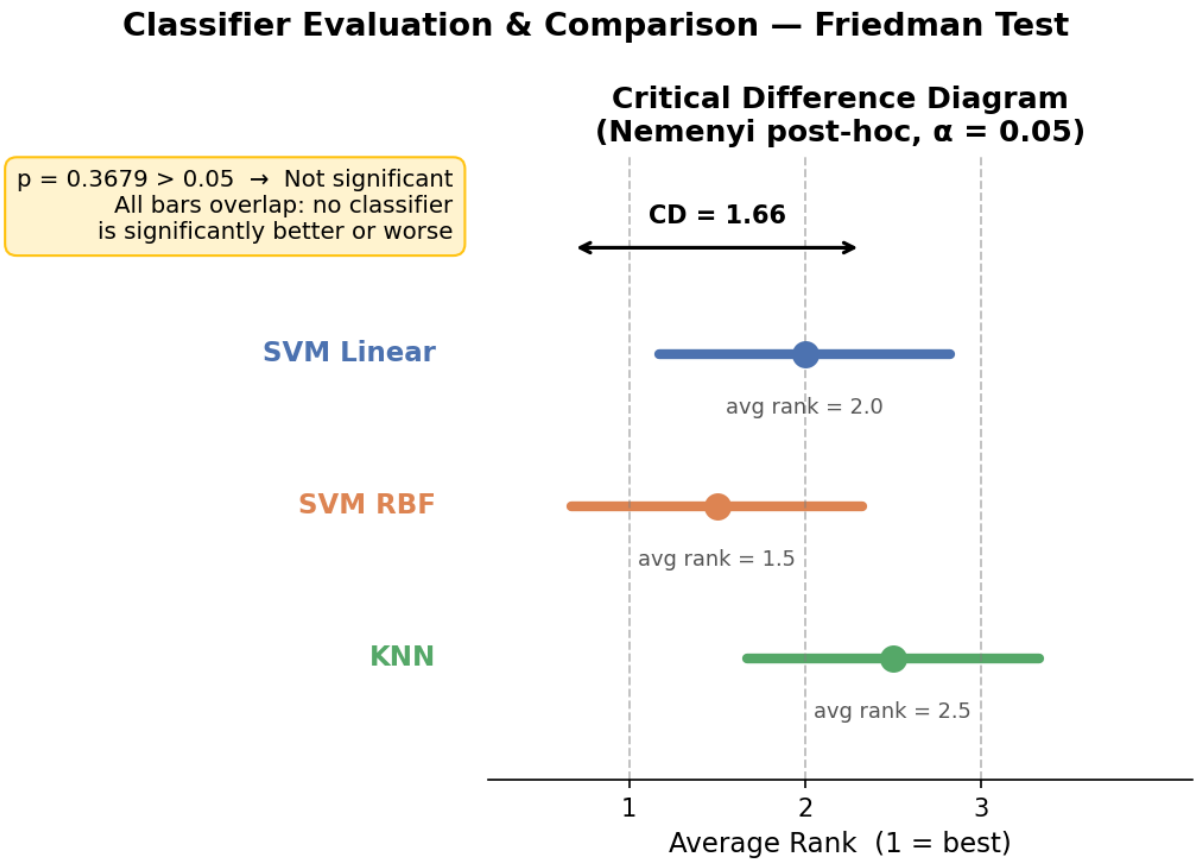

Figure 2: Critical difference diagram for the three classifiers based on average ranks over the four datasets.

4.3 Rank Matrix and Predominant Classifiers

Table 2 reports the rank matrix used in the Friedman test. For each dataset, the classifiers are ranked according to their accuracy (rank 1 = best), and an average rank over the four datasets is also shown. The classifier with rank 1 on a given dataset can be regarded as the *predominant* classifier on that dataset.

Classifier	DS1	DS2	DS3	DS4	Avg Rank
SVM Linear	1	2	3	2	2.00
SVM RBF	1	2	3	1	1.50
KNN	3	1	2	2	2.50

Table 2: Ranks (1 = best) of each classifier on each dataset and corresponding average ranks used in the Friedman test.

5 Discussion

This report compared SVM Linear, SVM RBF, and KNN on four datasets using the 5×2 cross-validation procedure and the Friedman test. The main empirical findings can be summarised as follows:

- **SVM RBF** achieves the best average rank (1.5) and is the most robust classifier across the four datasets.

- **SVM Linear** performs perfectly on linearly separable data (e.g., Dataset 1) but fails on non-linearly separable data (Datasets 3 and 4), where its accuracy is clearly inferior.
- **KNN** with $k = 5$ is competitive and reaches high accuracy on most datasets, but its average rank (2.5) is slightly worse than that of SVM RBF.
- The Friedman test finds no statistically significant difference among the classifiers, with a p-value of approximately 0.37 ($p = 0.3679 > 0.05$). This lack of significance is likely due to the limited number of datasets ($N = 4$), which reduces the power of the test.

Even though the average ranks suggest that SVM RBF is the best-performing and most stable classifier across the four problems, the statistical analysis indicates that these differences cannot be claimed as significant at the 5% level.

6 Conclusion

In this supervised learning assignment, I implemented and compared three classifiers (SVM with linear kernel, SVM with RBF kernel, and KNN) on four synthetic datasets using the 5×2 cross-validation procedure. A Friedman test with Nemenyi post-hoc analysis was then used to assess statistical significance.

The experimental results indicate that, at the 5% significance level, there is no strong evidence that any of the considered classifiers systematically outperforms the others across the four datasets; the Friedman test does *not* reject the null hypothesis of equal performance. Nevertheless, from a practical point of view and based on the observed average ranks and accuracies, SVM RBF appears to be the most robust choice, especially when the nature of the decision boundary (linear vs. non-linear) is unknown, as it generalises well to both cases. This highlights the importance of combining accuracy estimates with appropriate statistical tests when comparing models, rather than relying solely on raw accuracy values, and of clearly distinguishing between statistical significance and practical recommendation.